

Difference between JPA, Hibernate and Spring Data JPA

JPA (Java Persistence API)

- JPA is a Java Persistence API (JSR 338) specification.
- It defines standards for persisting, reading and managing Java objects.
- JPA does not provide its own implementation.
- Hibernate is one of the implementations of JPA.

Hibernate

- Hibernate is an ORM (Object Relational Mapping) framework.
- It implements the JPA specification.
- It maps Java objects to database tables.
- It provides Session, SessionFactory and Transaction APIs.

Spring Data JPA

- Spring Data JPA is not a JPA implementation.
- It is an abstraction over JPA implementations such as Hibernate.
- It reduces boilerplate code.
- It provides CRUD methods like save(), findById(), findAll() and deleteById().
- It manages transactions using Spring.

Feature	JPA	Hibernate	Spring Data JPA
Definition	Specification	ORM Framework	Spring Module
Implementation	No	Yes	No
Purpose	Defines persistence standard	Implements JPA	Reduces boilerplate code
Database Operations	Uses JPA implementation	Uses Session API	Uses JpaRepository
Transaction	Not directly	Manual/Spring	Managed by Spring

Conclusion

JPA is a specification, Hibernate is an implementation of JPA, and Spring Data JPA is a higher-level abstraction that simplifies database operations by reducing boilerplate code.